# Weaver Goldman

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## Education

**Worcester Polytechnic Institute** — *Projected* May 2027
*MS Data Science* — *GPA **4.0**/4.0*
*BS Computer Science*

**Courses**: MLOps, Machine Learning, Statistical Methods for ML, Database Management Systems, Algorithms

## Experience

**AI Engineering Intern** | *Regeneron Pharmaceuticals, Inc.* — May 2026 – Aug 2026
- Built an auditable AI/ML observability platform on AWS monitoring 30+ models and tens of thousands of model events, adopted by 3 pilot teams, and informed by 13 stakeholder interviews.
- Developed an environment-agnostic Python SDK using asyncio for scalable data ingestion and model analysis.
- Built a React dashboard to visualize model performance, data drift, and benchmarking results.
- Implemented an adaptive LLM benchmarking system with **>60%** lower API costs than a naive evaluation strategy

**Software Engineering Intern** | *Medtronic (Minimally Invasive Therapies Group)* — June 2025 – August 2025
- Saved **400+ hours** of manual work annually by automating surgical robot log analysis tasks with AI (LLMs).
- Developed an AI-powered tool to automatically analyze robot logs, performing expert-level analysis in minutes.
- Coordinated numerous AI agents and data sources with LangChain/LangGraph to perform complex tasks.

**Web Development Lab Assistant** | *Laboratory of Spaceflight and Planetary Exploration* — August 2024 – Present
- Under contract with **NASA**, developed a data-driven space mission planning web application.
- Optimized the database and graphing system to support more than **a million** different data points interactively.
- Led the development of an interactive, 3D Mars lander simulation.
- Built a robust CI/CD pipeline for the lab with Docker, unit testing, linting, and GitHub Actions.

**Software Development and Research Intern** | *Gittlen Cancer Research Labs* — May 2024 – August 2024
- Developed an Electron desktop application for extracting nerves and blood vessels from cloud-hosted medical scans, which reduces file sizes by up to **69,355%** (from 132.08 GB to 190.44 MB).
- Achieved approx. **4,000%** computation speed improvement with multithreading and multiprocessing.
- Utilized Docker to create a cross-platform Python backend and an extensible plugin system.

## Projects

**Scaling Diffusion Models to Large Sparse Graphs** | *PyTorch, Python* — August 2025 – May 2026
- 4-person senior project on efficient diffusion architectures for molecule and protein generation in drug discovery.
- Achieved linear space complexity for our diffusion process on large, sparse graphs.

**#TeamGraduate AI** | *Python, FastAPI, React, WordPress, Docker, Google Cloud Run* — Jan. 2026 – May 2026
- Developed a cloud-hosted LLM platform for generating curriculum-aligned (Namibian) educational content.
- Built AI authoring tools embedded in WordPress that reduced lesson creation time by **87.7%**.

**Identifying Transit Accessibility Gaps in NYC** | *Scikit-Learn, Python, Plotly, GTFS* — April 2025
- Identified underserved areas in NYC by using distance to the nearest subway station as a proxy for accessibility.
- Predicted weekly ridership revenue per location using a Random Forest Regressor ($R^2 = 0.675$), based on **6** years of data.
- Visualized results with interactive geospatial maps, highlighting areas of opportunity based on accessibility and predicted revenue metrics (potential stops predicted to produce **>$4 million** in revenue annually).

## Skills

**Languages**: Python, JavaScript/TypeScript, SQL, C/C++, Java, LaTeX
**Libraries**: PyTorch, LangChain, TensorFlow, PostgreSQL, React, Angular, Streamlit, Plotly Dash, Three.js
**DevOps/MLOps**: GCP, Azure, AWS, Cloudflare, Prometheus, Grafana, Bash, Weights and Biases
**Tools/Frameworks**: Git/GitHub, Jira, Docker, Linux, Figma